Concept-Grounded Image Captioning for Fine Art Using CLIP Representations

Impact Statement

This work addresses the problem of generating captions for artworks using a lightweight and interpretable framework. Unlike many existing approaches that rely on large multimodal architectures and expensive training procedures, the proposed method separates visual understanding from language generation through an intermediate concept prediction stage. This design improves interpretability while keeping the training pipeline computationally efficient.

The study demonstrates that pretrained vision–language representations can support artwork caption generation even in limited-resource research settings. By combining CLIP-based visual embeddings with concept-guided caption generation, the framework provides a practical direction for fine art understanding without requiring large-scale retraining.

The proposed approach is relevant for research areas involving vision–language learning, digital art analysis, cultural heritage understanding, and low-resource multimodal AI systems. The work also highlights the importance of interpretable intermediate representations in image captioning systems.